

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	Easy

Question

Which expression is equivalent to $(2x^2 + x - 9) + (x^2 + 6x + 1)$?

Answer

- A. $2x^2 + 7x + 10$
- B. $2x^2 + 6x - 8$
- C. $3x^2 + 7x - 10$
- D. $3x^2 + 7x - 8$

Correct Answer: D

Rationale

Choice D is correct. The given expression is equivalent to $(2x^2 + x + (-9)) + (x^2 + 6x + 1)$, which can be rewritten as $(2x^2 + x^2) + (x + 6x) + (-9 + 1)$. Adding like terms in this expression yields $3x^2 + 7x + (-8)$, or $3x^2 + 7x - 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.